

NAME : _____

CLASS : _____



JURONG PIONEER JUNIOR COLLEGE
JC2 Preliminary Examination 2022

BIOLOGY
Higher 2

9744/03
21 September 2022

Paper 3 Structured and Free-response Questions

2 hours

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your class and name in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
Section B	
Total	

This document consists of **20** printed pages.

Question 1 starts on page 3

Section A

Answer **all** the questions in this section.

- 1** Colorectal cancer and hepatocellular carcinoma (HCC) are amongst the leading causes of cancer-related deaths worldwide. Colorectal cancer is the cancer of the colon or rectum. It is amongst the top three causes of cancer in both men and women globally, with more than a million people diagnosed with colorectal cancer each year.

Most colorectal cancers are due to old age and lifestyle factors, with only a small number of cases due to underlying genetic factors.

- (a)** State two environmental causes of cancer.

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- (b) Colorectal cancer often begins as an adenoma – a benign tumour that develops at the inner lining of the large intestine. Though benign, Fig. 1.1 shows how adenomas may transform to become malignant as a result of genetic alterations over time.

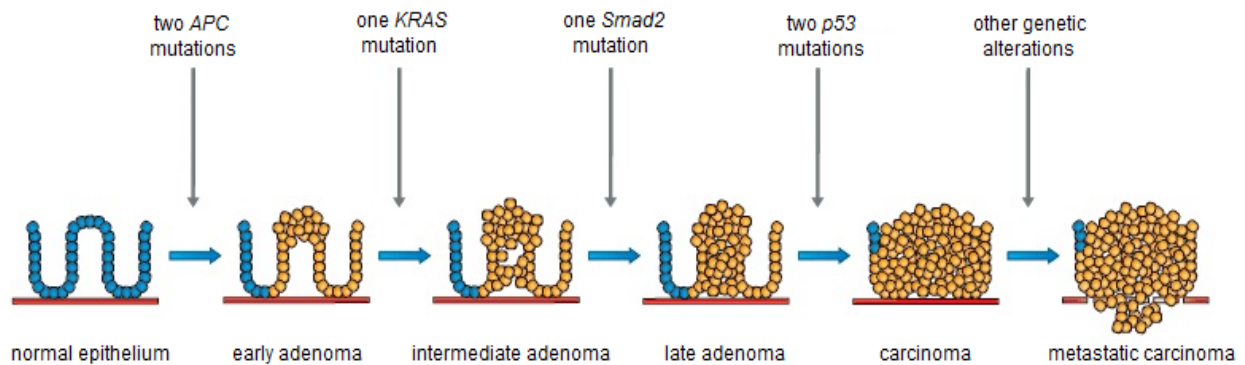


Fig. 1.1

Explain how Fig. 1.1 provides evidence for the development of cancer as a multi-step process.

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- (c) Different types of gene mutation have been found to affect tumour suppressor genes.

Explain why a base pair deletion in a tumour suppressor gene is more likely to produce a non-functional protein than a base pair substitution.

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Unlike colorectal cancer, the incidence of HCC can be attributed to different genetic and epigenetic alterations. Epigenetics refers to heritable states of gene expression without alteration to the DNA sequence itself. Epigenetic changes such as DNA hypermethylation or hypomethylation, dysregulation of histone modification patterns, chromatin remodelling and aberrant expression of micro-RNAs (miRNAs) are associated with HCC.

miRNAs are non-coding single-stranded RNAs that are typically 19 to 25 nucleotides in length. They mostly function as negative gene regulators by binding to their complementary sequence located at the 3' untranslated region of their target mRNA, thereby forming miRNA-mRNA duplexes and silencing gene expression.

- (d) State the level at which miRNAs regulate gene expression.

..... [1]

- (e) Fig. 1.2 shows how chronic liver damage and the replacement of normal liver tissue by fibrous scar tissue can lead to liver cirrhosis. Cirrhosis is generally irreversible once it occurs; more often than not, it leads to HCC.

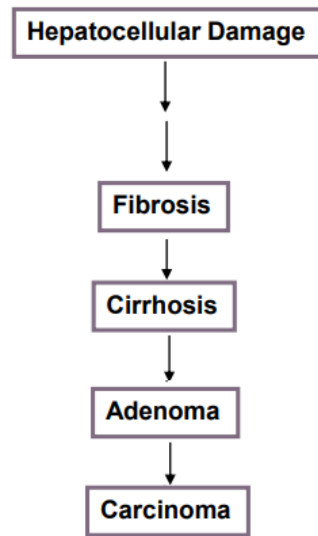


Fig. 1.2

A study investigated the association between differential levels of miRNA expression in liver cirrhosis. Non-cirrhotic and cirrhotic liver samples were used and the results are shown in Fig. 1.3.

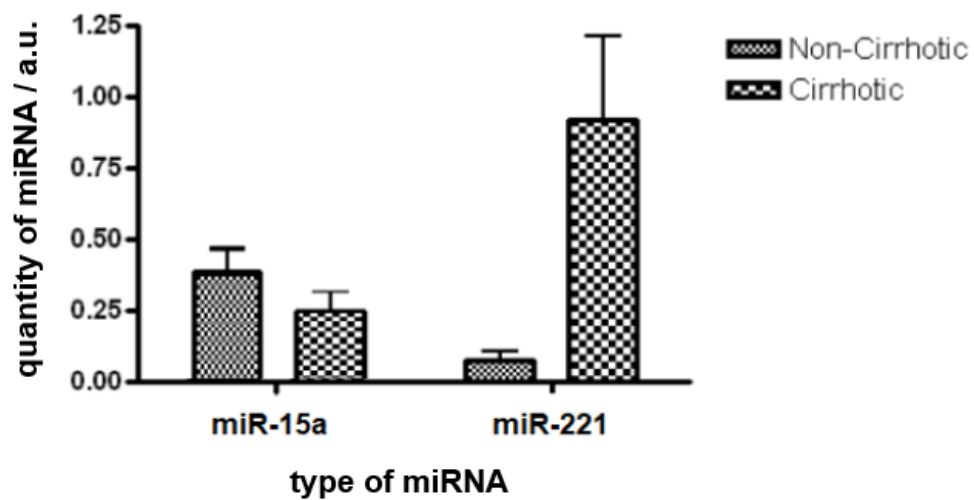


Fig. 1.3

- (i) Compare the effect of the relative quantities of miRNA in leading to liver cirrhosis.

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- (ii) 'oncomiRs' are miRNAs that promote oncogenesis by negatively regulating important tumour suppressor genes. On the other hand, 'anti-oncomiRs' are miRNAs that exert tumour-suppressive effects by repressing oncogenes.

Deduce if miR-15a and miR-221 is an 'oncomiR' or 'anti-oncomiR' and justify your answer.

miR-15a

justification

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miR-221

justification

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[4]

- (f) The molecular genetics of HCC have recently been extensively characterised. One of the most consistent epigenetic changes in HCC is that of DNA hypermethylation at the promoters of target genes.

- (i) Describe how DNA hypermethylation at promoters can alter expression levels of the target genes.

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- (ii) The *p16INK4a* gene is located on chromosome 9 and is one of the most frequently altered genes observed in HCC. The prevalence of hypermethylation of its promoter in various types of liver tissues are shown in Tables 1.1 and 1.2.

Table 1.1

sample	percentage of samples which showed <i>p16INK4a</i> promoter hypermethylation
HCC	54.5
non-tumourous	15.6

Table 1.2

sample	percentage of samples which showed <i>p16INK4a</i> promoter hypermethylation
cirrhotic	22.7
non-cirrhotic	9.2

Predict whether *p16INK4a* is an oncogene or tumour suppressor gene and suggest a possible role of its protein product in the cell cycle.

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- (g) Another common hallmark of HCC is that of telomerase reactivation.

A student made the following claim:

“In eukaryotes, telomerase synthesises additional lengths of DNA that are added to the ends of chromosomes, thus solving the end-replication problem that results from each round of DNA replication.”

- (i) Account for the end-replication problem.

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- (ii) Evaluate the validity of the student's claim that the addition of DNA to chromosomal ends solves the end-replication problem.

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- (iii) Telomerase is **not** present in prokaryotic cells. Suggest why prokaryotes do **not** have telomerase.

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[Total: 30]

- 2 Gene expression in eukaryotes is regulated at different levels and involves various regulatory proteins and enzymes. At translational level of regulation, regulatory proteins such as translation initiation factors and translational repressors are involved.

(a) Describe how gene expression may be regulated by translational repressors.

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Another regulatory protein, ubiquitin, is a highly conserved protein comprising 76 amino acids, ubiquitously expressed in all tissues in eukaryotes.

(b) State the role of ubiquitin in regulating gene expression in eukaryotes.

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Fig. 2.1 shows the structure of an ubiquitin molecule.

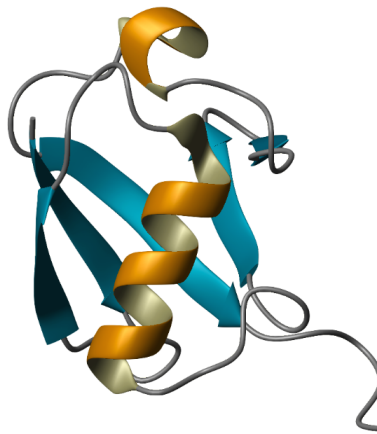


Fig 2.1

(c) Describe **two** similarities between the secondary structures found in ubiquitin.

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Ubiquitin is synthesised at cytosolic ribosomes. mRNA, tRNA and rRNA are involved in the synthesis of ubiquitin.

(d) Describe the roles of tRNA in the synthesis of ubiquitin.

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During protein synthesis, transcription precedes translation. Researchers measured the error rates during transcription and translation in yeast cells. The results are shown in Table 2.1.

Table 2.1

process	error rate
transcription	10^{-5} per base
translation	10^{-3} per codon

(e) Suggest why such a difference in error rates during transcription and translation is tolerated by the yeast cell.

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[Total: 10]

- 3 Algae are aquatic photosynthetic protists. Some researchers genetically modified the unicellular alga, *Chlorella vulgaris*, to try to increase the rate of the light independent stage of photosynthesis.

C. vulgaris was modified to increase the expression of the gene coding for aldolase. Aldolase is an enzyme that causes an increase in the concentration of rubisco.

Two cultures of *C. vulgaris*, one that was not genetically modified (unmodified) and one genetically modified, were grown under controlled conditions for 14 days. Samples were taken from the cultures at regular intervals during the 14 days to obtain measurements of dry mass. The results are shown in Fig. 3.1.

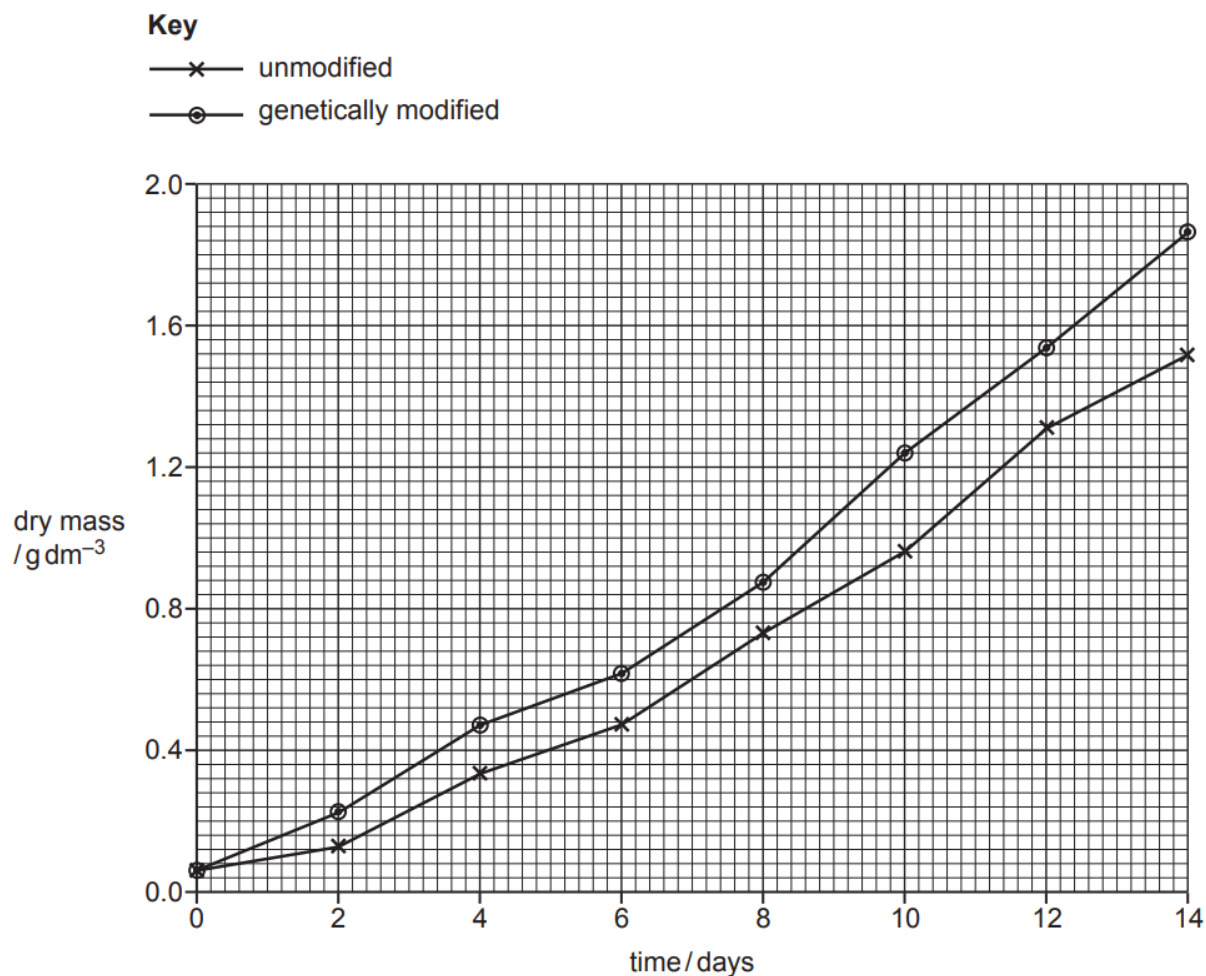


Fig. 3.1

- (a) With reference to Fig. 3.1, describe the differences between the results for the two cultures.

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- (b) Explain how the Calvin cycle was affected by the genetic modification of *C. vulgaris*.

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- (c) Intermediate products of the Calvin cycle are needed to produce organic molecules for use by the cell. One such organic molecule is phospholipid, a major constituent of cell membranes.

Outline the functions of membranes within cells.

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- (d) Planting large numbers of trees is one way to reduce global atmospheric CO₂ concentration. Large scale culture of genetically modified *C. vulgaris* could also reduce global atmospheric CO₂ concentration.

Suggest one advantage of using genetically modified *C. vulgaris* instead of trees to reduce global atmospheric CO₂ concentration.

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..... [1]

[Total: 10]

NAME : _____

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Section BAnswer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)**, as indicated in the question.

- 4 (a) Describe how transformation, transduction and conjugation give rise to variation in prokaryotic genomes and explain why genetic variation is important in bacteria. [15]
- (b) Describe the structures and organisation of viral, prokaryotic and eukaryotic genomes. [10]

[Total: 25]

- 5 (a) Outline and contrast chemiosmosis in photosynthesis and respiration. [15]
- (b) Describe how the molecular structure and properties of cellulose are related to its functions. [10]

[Total: 25]

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